

UNIVERSITI KEBANGSAAN MALAYSIA
The National University of Malaysia

Carbon Nanotubes: Structure and Band Structure

KKKC6353 Nanoelectronic Devices — Lecture 4, Document 1 of 4

Dr Muhammad Faiz Bukhori
Universiti Kebangsaan Malaysia

October 2026

Table of Contents

The argument of this document **3**

1 Carbon bonding and the graphene lattice **4**

1.1 sp^2 bonding 4

1.2 The honeycomb as a lattice with a basis 4

2 The π bands of graphene **5**

2.1 The tight-binding model 5

2.2 The Dirac cone 6

3 Rolling up: the chiral vector **7**

3.1 Diameter, chiral angle and unit cell 7

3.2 Single-, double- and multi-walled tubes 8

4 Zone folding: metal or semiconductor **9**

4.1 The boundary condition 9

4.2 Through K or not 9

4.3 The band gap 10

4.4 Corrections for real tubes 11

5 Subbands, effective mass and density of states **12**

5.1 Subbands and effective mass 12

5.2 The density of states 12

5.3 Identifying tubes by their optical transitions 13

6 Transport **14**

6.1 The Landauer conductance 14

6.2 Scattering and current saturation 14

6.3 Thermal and mechanical properties 15

7 From physics to material: growth, sorting and alignment **16**

7.1 Growth 16

7.2 Sorting 17

7.3 Alignment and density 17

8 Summary **17**

Corrections to the old notes 18

Words to know 18

In-class animation **18**

Sources and further reading **19**

The argument of this document

> **Recap from Lecture 2**
>
> A carbon nanotube is graphene rolled into a seamless cylinder, about 1 nm across and up to centimetres long (Iijima, 1991).
>
> How the sheet is rolled, labelled by two integers (n, m), decides whether the tube is a metal or a semiconductor. Statistically one in three is metallic.
>
> Transport in short tubes is ballistic, but even a perfect tube has a resistance of about 6.5 kΩ, set by its number of channels (Lecture 2, Document 2, Section 5).

Lecture 2 introduced the carbon nanotube as a sheet of graphene rolled into a tube: about one tube in three is metallic, the band gap of the others shrinks as the tube gets wider, and electrons can cross a short tube without scattering. Those statements are true, but they were given as facts. This document **derives** them, because the transistor physics of Document 2 depends on the details: the size of the gap sets the off-current, the shape of the bands sets the effective mass and the quantum capacitance, and the number of conducting channels sets the minimum resistance.

The argument has four steps, and each one is a short calculation:

1. The π electrons of graphene are described well by a **nearest-neighbour tight-binding model** with one parameter, $\gamma_0 \approx 2.7$ eV. It gives two bands that touch at six points, the **K points**, where the dispersion is a cone. Graphene is a semiconductor with zero gap.
2. Rolling the sheet imposes a **periodic boundary condition** around the circumference. Only lines of allowed wavevectors survive, spaced $2/d$ apart.
3. If a line passes through a K point the tube is **metallic**; if not, the nearest line misses K by exactly one third of the spacing, and the tube has a gap $E_g = 2a_{cc}\gamma_0/d$. Which case occurs depends only on $(n - m)$ mod 3.
4. Each line is a one-dimensional **subband**. One-dimensional subbands have a density of states with sharp peaks, a hyperbolic dispersion with a small effective mass, and a conductance quantum of $4e^2/h$.

The document ends with what this means for making devices: how tubes are grown, sorted and lined up in 2026, and what is still missing.

Notation: $a_{cc} = 0.142$ nm is the carbon–carbon bond length; $a = \sqrt{3}\, a_{cc} = 0.246$ nm is the lattice constant of graphene; d is the tube diameter; $\hbar v_F$ is the slope of the Dirac cone. Energies are measured from the Fermi level of undoped graphene.

1 Carbon bonding and the graphene lattice

> **Key ideas**
> In graphene each carbon uses three sp^2 orbitals for strong in-plane σ bonds. The fourth electron sits in a p_z orbital, and these p_z orbitals form the π bands that carry current.
> The honeycomb is a triangular lattice with two atoms per unit cell, A and B. That two-atom basis is what gives two π bands.

1.1 sp^2 bonding

A carbon atom has four valence electrons, in the $2s$, $2p_x$, $2p_y$ and $2p_z$ orbitals. In graphene, three of them mix into three **sp^2 hybrid** orbitals in the plane, 120° apart, which form σ bonds with the three nearest neighbours. The σ bonds are among the strongest in nature, which is why graphene and nanotubes are so stiff (Young's modulus about 1 TPa) and so resistant to electromigration. The σ bands lie several electronvolts away from the Fermi level and play no part in conduction.

The fourth electron occupies the $2p_z$ orbital, perpendicular to the sheet. Neighbouring p_z orbitals overlap side-on to form delocalised π bonds. **All the low-energy electronic properties come from these π electrons**, one per atom. In a nanotube the sheet is curved, which mixes a little σ character into the π orbitals. The mixing matters only for very thin tubes ($d < 0.8$ nm) and for the very small gaps discussed in Section 4.4.

1.2 The honeycomb as a lattice with a basis

The honeycomb is *not* a Bravais lattice: the two atoms in Figure 1.1 marked A and B have differently oriented neighbours. It is a triangular lattice with lattice vectors

$$\mathbf{a}_1 = a\left(\sqrt{3}/2\,,\ 1/2\right), \qquad \mathbf{a}_2 = a\left(\sqrt{3}/2\,,\ -1/2\right), \qquad a = |\mathbf{a}_1| = \sqrt{3}\,a_{cc} = 0.246\ \text{nm}$$

and a two-atom basis. Every A atom has three B neighbours at distance $a_{cc} = 0.142$ nm, and vice versa. The reciprocal lattice is also triangular, with $|\mathbf{b}_1| = |\mathbf{b}_2| = 4\pi/\left(\sqrt{3}a\right)$, and the first Brillouin zone is a hexagon. Its six corners are the **K points**, at a distance $4\pi/(3a) = 17.0\ \text{nm}^{-1}$ from the centre Γ. Only two of the six are inequivalent; they are called K and K′.

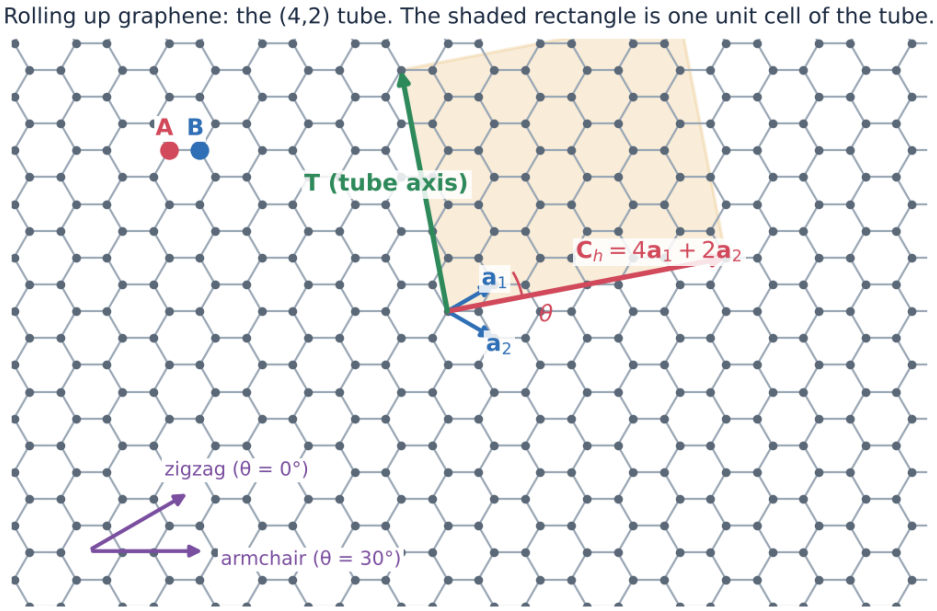


Figure 1.1. *The graphene lattice, with the lattice vectors* $\mathbf{a}_1$ *and* $\mathbf{a}_2$*, the chiral vector* $\mathbf{C}_h$ *of a (4,2) tube and the translation vector* $\mathbf{T}$ *along the tube axis. Rolling the sheet so that the two ends of* $\mathbf{C}_h$ *meet makes the tube; the shaded rectangle becomes its unit cell. The chiral angle* θ *is measured from the zigzag direction. Figure made for this course.*

Check your understanding

1. Why do the σ bands play no part in electrical conduction in graphene?
2. Why does the honeycomb need a two-atom basis, and what is the consequence for the number of π bands?

2 The π bands of graphene

> **Key ideas**
>
> A tight-binding model with one hopping energy γ_0 gives $E(\mathbf{k}) = \pm\gamma_0|f(\mathbf{k})|$.
>
> The two bands touch at the K points. Near K the dispersion is a cone, $E = \pm\hbar v_F|\boldsymbol{\kappa}|$, with $v_F \approx 8.7 \times 10^5$ m/s.
>
> Undoped graphene has its Fermi level exactly at the touching point: a zero-gap semiconductor.

2.1 The tight-binding model

Write the π wavefunction as a combination of Bloch sums of p_z orbitals on the A and B sublattices, $\psi = c_A u_A + c_B u_B$. Substituting into $H\psi = E\psi$, keeping only hopping between nearest neighbours, and neglecting the overlap between orbitals on different atoms gives a 2 × 2 eigenvalue problem:

$$\begin{pmatrix} 0 & -\gamma_0 f(\mathbf{k}) \\ -\gamma_0 f^*(\mathbf{k}) & 0 \end{pmatrix} \begin{pmatrix} c_A \\ c_B \end{pmatrix} = E \begin{pmatrix} c_A \\ c_B \end{pmatrix}, \qquad f(\mathbf{k}) = 1 + e^{i\mathbf{k}\cdot\mathbf{a}_1} + e^{i\mathbf{k}\cdot\mathbf{a}_2}$$

The diagonal elements (the on-site energy) are set to zero, which fixes the energy zero. The off-diagonal element is the sum of hops to the three neighbours, each weighted by its Bloch phase. The eigenvalues are

$$E_{\pm}(\mathbf{k}) = \pm\gamma_0 \sqrt{1 + 4\cos\frac{\sqrt{3}k_x a}{2}\cos\frac{k_y a}{2} + 4\cos^2\frac{k_y a}{2}}$$

The minus sign is the filled **π (bonding) band**; the plus sign is the empty **π* (antibonding) band**. With one π electron per atom, the π band is exactly full. The hopping energy γ_0 is fitted to experiment; values between 2.5 and 3.0 eV are used in the literature. This document uses $\gamma_0 = 2.7$ eV throughout. Including the overlap integral makes the two bands asymmetric, but changes nothing below.

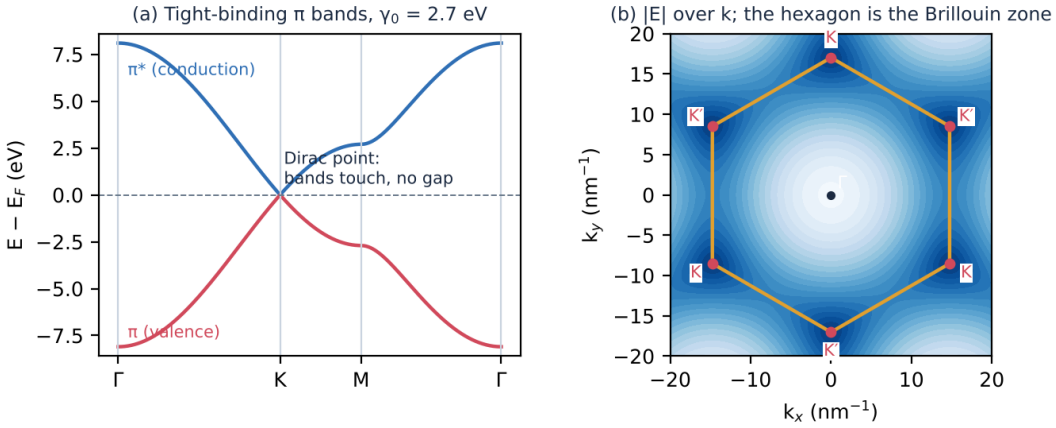


Figure 1.2. *(a) The π bands of graphene along the high-symmetry path Γ–K–M–Γ. The bands touch at K. (b) Contours of $|E|$ over the (k_x, k_y) plane. The bands touch at the six corners of the hexagonal Brillouin zone, of which two (K and K′) are inequivalent. Figure made for this course from the tight-binding expression above, $\gamma_0 = 2.7$ eV.*

2.2 The Dirac cone

At a K point, $f(\mathbf{K}) = 0$: the three Bloch phases add to zero, and both bands are at $E = 0$. Expanding f to first order in the distance $\boldsymbol{\kappa} = \mathbf{k} - \mathbf{K}$ gives

$$E_{\pm} = \pm\hbar v_F |\boldsymbol{\kappa}|, \qquad \hbar v_F = 3/2\ a_{cc}\,\gamma_0$$

The dispersion near each K point is a **cone**, the same in every direction. Electrons there behave like massless particles moving at a fixed speed v_F, whatever their energy. Within about 1 eV of the Fermi level this is the only part of the band structure that matters, and every result in Sections 4 and 5 follows from it.

> **Worked example: the Fermi velocity**
>
> $\hbar v_F = 1.5 \times 0.142$ nm $\times$ 2.7 eV $= 0.575$ eV nm. With $\hbar = 6.58 \times 10^{-16}$ eV s, $v_F = 0.575 \times 10^{-9} / 6.58 \times 10^{-16} = 8.7 \times 10^5$ m/s, about 1/340 of the speed of light. Measurements on graphene give $v_F \approx 1 \times 10^6$ m/s, which corresponds to $\gamma_0 \approx 3.1$ eV. This spread in γ_0 is why textbooks quote slightly different band-gap formulas (Section 4.3).

Check your understanding

1. Why does $f(\mathbf{k})$ vanish at the K points? (Hint: three unit vectors 120° apart.)
2. Where does the Fermi level of undoped graphene sit, and why?

3 Rolling up: the chiral vector

> **Key ideas**
> A tube is fixed by its chiral vector $\mathbf{C}_h = n\mathbf{a}_1 + m\mathbf{a}_2$, which becomes the circumference.
> From (n, m) follow the diameter, the chiral angle, the length of the unit cell along the axis, and the number of atoms in that cell.
> Zigzag tubes are $(n, 0)$, armchair tubes (n, n); everything else is chiral.

3.1 Diameter, chiral angle and unit cell

Rolling the sheet so that two lattice points coincide makes a seamless tube. The vector joining them, $\mathbf{C}_h = n\mathbf{a}_1 + m\mathbf{a}_2$ with integers $0 \le m \le n$, becomes the circumference:

$$|\mathbf{C}_h| = a\sqrt{n^2 + nm + m^2}, \qquad d = \frac{|\mathbf{C}_h|}{\pi} = \frac{a}{\pi}\sqrt{n^2 + nm + m^2}$$

The **chiral angle** θ is the angle between $\mathbf{C}_h$ and $\mathbf{a}_1$ (the zigzag direction):

$$\cos\theta = \frac{2n + m}{2\sqrt{n^2 + nm + m^2}}, \qquad 0^\circ \le \theta \le 30^\circ$$

The tube axis is perpendicular to $\mathbf{C}_h$. The shortest lattice vector along it is the **translation vector**

$$\mathbf{T} = t_1\mathbf{a}_1 + t_2\mathbf{a}_2, \qquad t_1 = \frac{2m + n}{d_R}, \quad t_2 = -\frac{2n + m}{d_R}, \quad |\mathbf{T}| = \frac{\sqrt{3}\,|\mathbf{C}_h|}{d_R}$$

where $d_R = \gcd(2m + n, 2n + m)$. The rectangle spanned by $\mathbf{C}_h$ and $\mathbf{T}$ is the tube's unit cell. It contains $N = 2(n^2 + nm + m^2)/d_R$ graphene unit cells, so $2N$ carbon atoms.

The three families are drawn in Figure 1.3:

- **Zigzag** $(n, 0)$, $\theta = 0^\circ$: the circumference follows a zigzag chain of bonds.
- **Armchair** (n, n), $\theta = 30^\circ$: the circumference follows an armchair-shaped chain.
- **Chiral**, $0^\circ < \theta < 30^\circ$: the hexagons spiral around the axis. A chiral tube and its mirror image are different molecules (left- and right-handed), which matters for sorting but not for the band structure.

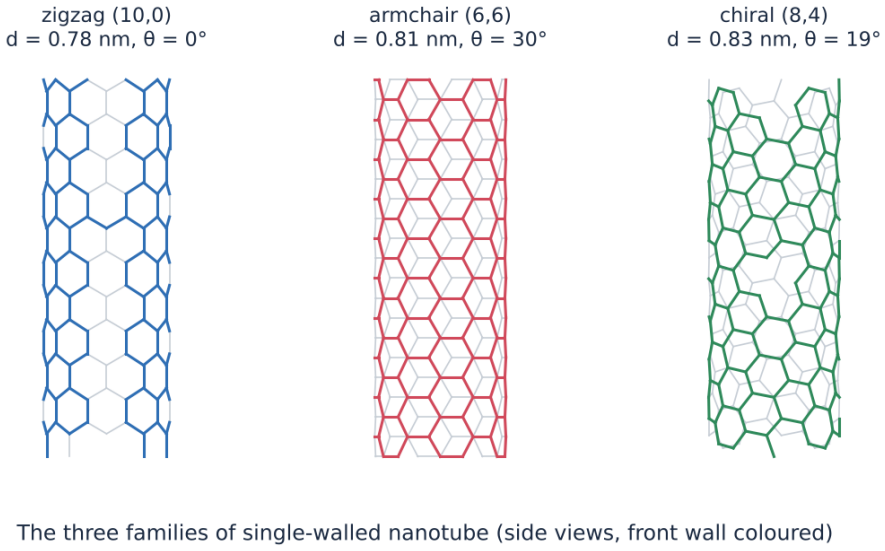


Figure 1.3. *Three single-walled nanotubes of similar diameter, drawn by rolling the lattice of Figure 1.1. The front wall is coloured and the back wall is grey. Figure made for this course.*

> **Worked example: three tubes**
>
> **(10,0):** $|\mathbf{C}_h| = 0.246\sqrt{100} = 2.46$ nm, so $d = 0.783$ nm; $d_R = \gcd(10,20) = 10$, $|\mathbf{T}| = \sqrt{3} \times 2.46/10 = 0.426$ nm; $N = 2 \times 100/10 = 20$, so 40 atoms per cell.
>
> **(6,6):** $|\mathbf{C}_h| = 0.246\sqrt{108} = 2.56$ nm, $d = 0.814$ nm, $\theta = 30°$, $|\mathbf{T}| = a = 0.246$ nm, 24 atoms per cell.
>
> **(8,4):** $|\mathbf{C}_h| = 0.246\sqrt{112} = 2.60$ nm, $d = 0.829$ nm, $\cos\theta = 20/\left(2\sqrt{112}\right)$, so $\theta = 19.1°$; $d_R = \gcd(16,20) = 4$, $|\mathbf{T}| = 1.13$ nm, $N = 56$, so 112 atoms per cell.
>
> The three tubes have almost the same diameter, but the chiral tube's unit cell is nearly three times as long and holds three to five times as many atoms. That is why chiral tubes are expensive to simulate atom by atom, and why device simulators often use zigzag tubes as stand-ins.

3.2 Single-, double- and multi-walled tubes

A **single-walled** nanotube (SWNT) is one rolled sheet. Electronics uses SWNTs, typically 0.8–2 nm in diameter. A **multi-walled** tube (MWNT) is several concentric tubes about 0.34 nm apart, the spacing between layers in graphite; a double-walled tube (DWNT) is the two-wall case. In an MWNT the walls have different chiralities, so some are metallic and some semiconducting; with outer diameters of 5–50 nm, even the semiconducting walls have gaps too small to switch off at room temperature. MWNTs are therefore used as conductors (interconnects, electrodes, composites) and SWNTs as transistor channels.

Check your understanding

1. Compute d and θ for a (13,0) tube and a (7,7) tube.
2. Why is an MWNT of 20 nm diameter no use as a transistor channel at room temperature?

4 Zone folding: metal or semiconductor

> **Key ideas**
> Around the circumference the wavefunction must repeat, so $\mathbf{k} \cdot \mathbf{C}_h = 2\pi\mu$. The allowed states form parallel lines in the Brillouin zone, $2/d$ apart.
> If a line passes through K the tube is metallic. This happens when $n - m$ is divisible by 3.
> Otherwise the nearest line misses K by one third of the spacing, and $E_g = 2a_{cc}\gamma_0/d \approx 0.77$ eV nm$/d$.

4.1 The boundary condition

Along the axis the tube is long, so the wavevector $k_\parallel$ along $\mathbf{T}$ is effectively continuous. Around the circumference the wavefunction must return to itself after one turn:

$$\mathbf{k} \cdot \mathbf{C}_h = 2\pi\mu, \qquad \mu = 0,1, \dots, N-1$$

So the component of $\mathbf{k}$ along $\mathbf{C}_h$ is quantised in steps of $2\pi/|\mathbf{C}_h| = 2/d$. In the two-dimensional Brillouin zone of graphene the allowed states form N parallel **cutting lines**, each of length $2\pi/|\mathbf{T}|$ and spaced $2/d$ apart. The band structure of the tube is the graphene band structure sampled along these lines: each line gives one one-dimensional **subband** $E_\mu(k_\parallel)$. This construction is called **zone folding**.

4.2 Through K or not

Near the Fermi level only the cutting lines closest to K and K′ matter. There are two cases.

- **A line passes through K.** That subband contains the Dirac point, so it disperses linearly through $E = 0$ with no gap. The tube is a **metal**: more precisely, a one-dimensional conductor with two linear bands crossing at the Fermi level.
- **No line passes through K.** The position of K relative to the lines is then fixed by symmetry: the nearest line misses K by exactly **one third** of the spacing, on one side or the other. The tube is a **semiconductor**.

Working out which case occurs is a matter of checking whether $\mathbf{K} \cdot \mathbf{C}_h$ is a multiple of 2π. The result is the rule quoted in Lecture 2:

(n, m) is metallic if $(n - m)$ is divisible by 3; otherwise it is semiconducting.

All armchair tubes ($n = m$) are therefore metallic, one zigzag tube in three is metallic, and so is one chiral tube in three. **The rule is general; armchair tubes are only the most obvious case.** (Kaushik and Majumder's chapter, one of the old notes, attaches the rule to armchair tubes alone.) If all (n, m) were equally likely, one third of tubes would be metallic, which is close to what growth produces.

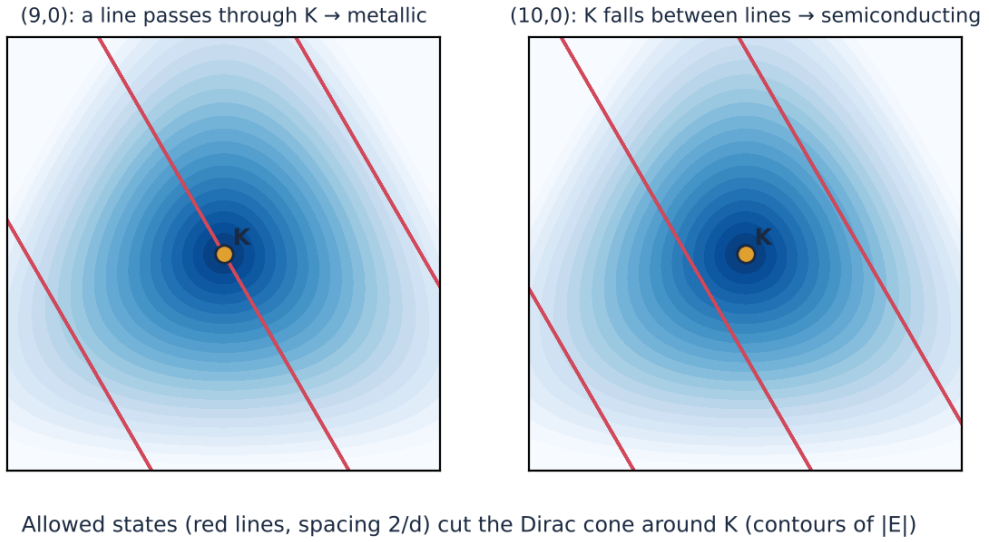


Figure 1.4. *Zone folding near a K point (gold). Contours show $|E|$ of graphene; red lines are the allowed wavevectors of the tube, spaced $2/d$ apart. For (9,0) a line passes through K and the tube is metallic. For (10,0) the nearest line misses K by one third of the spacing and a gap opens. Figure made for this course from the tight-binding model.*

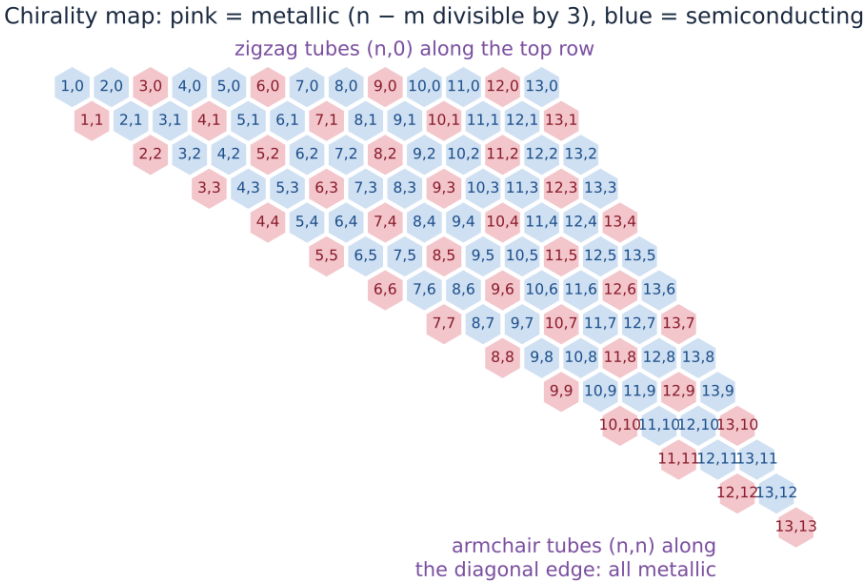


Figure 1.5. *The chirality map. Each hexagon is one (n, m); pink tubes are metallic. Along any row one tube in three is metallic, and every armchair tube on the diagonal edge is metallic. Figure made for this course.*

4.3 The band gap

The gap follows directly from the cone. The nearest cutting line misses K by

$$\Delta k = \frac{1}{3} \cdot \frac{2}{d} = \frac{2}{3d}$$

Along that line the energy is $E = \pm\hbar v_F \sqrt{\Delta k^2 + k_\parallel^2}$, which has its minimum $\pm\hbar v_F \Delta k$ at $k_\parallel = 0$. The gap is twice that:

$$E_g = 2\hbar v_F \Delta k = \frac{4\hbar v_F}{3d} \quad \Rightarrow \quad \boxed{E_g = \frac{2a_{cc}\gamma_0}{d} \approx \frac{0.77\ \text{eV nm}}{d}}$$

using $\hbar v_F = 3/2\, a_{cc}\gamma_0$ and $\gamma_0 = 2.7$ eV. Figure 1.6 checks this against the full tight-binding calculation for every tube between 0.6 and 3 nm: the formula is accurate to a few per cent above 1 nm.

A note on the prefactor. The literature quotes $E_g \approx 0.7/d$, $0.8/d$ or $0.9/d$ (eV, d in nm). These are the same formula with γ_0 between 2.5 and 3.1 eV. Lecture 2 used 0.8 and one of the old notes uses 0.7; this lecture uses 0.77. Measured optical gaps are further shifted by electron–electron interactions and excitons. When a number matters, state which γ_0 you used.

> **Worked example: the gaps of real device tubes**
>
> Transistor tubes are typically 1.2–1.8 nm across. For $d = 1.5$ nm, $E_g = 0.77/1.5 = 0.51$ eV, about half the gap of silicon. For $d = 1.0$ nm, $E_g = 0.77$ eV; for $d = 2.0$ nm, 0.38 eV. A tube wider than about 2 nm has a gap under 0.4 eV, and Document 2 will show that its off-current is then too high for logic. This is the first constraint on diameter. The second, from contacts, pushes the other way.

4.4 Corrections for real tubes

Zone folding treats the tube as a flat sheet with a boundary condition. Two effects of curvature change the answer slightly:

- **Small gaps in "metallic" tubes.** Curvature shifts the K point a little off the cutting line in every metallic tube except the armchair ones, opening a gap of order 10–100 meV that falls as $1/d^2$. Only armchair tubes are truly metallic. At room temperature the small-gap tubes still conduct like metals, which is what matters for devices.
- **σ–π mixing in very thin tubes.** Below about 0.8 nm the strong curvature mixes σ and π orbitals and pulls the bands down; some tubes predicted to be semiconducting become metallic. Tubes this thin are rare in devices.

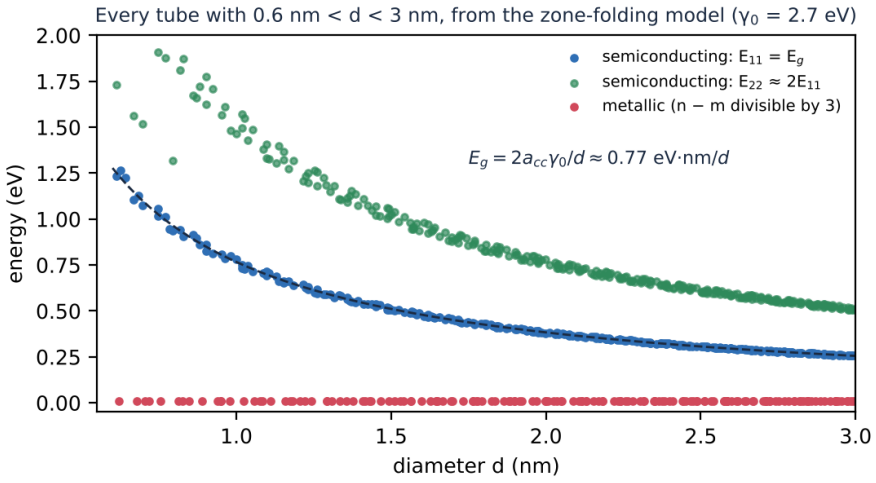


Figure 1.6. The band gap E_{11} and the next optical transition E_{22} of every tube with diameter between 0.6 and 3 nm, from the zone-folding model. Metallic tubes are at zero. The dashed line is $E_g = 2a_{cc}\gamma_0/d$. Figure made for this course.

Check your understanding

1. Classify (12,0), (12,1), (11,5) and (9,6) as metallic or semiconducting.
2. Estimate E_g for a (17,0) tube. Why does the zone-folding answer for a (5,0) tube not match experiment?

5 Subbands, effective mass and density of states

> **Key ideas**
>
> Each cutting line is a subband with a hyperbolic dispersion. Near its edge it looks parabolic, with an effective mass of about $0.09\,m_0/d(\text{nm})$.
>
> One-dimensional subbands have a density of states with sharp peaks at each band edge: the van Hove singularities.
>
> Optical transitions between the peaks (E_{11}, E_{22}, ...) are how individual (n, m) species are identified.

5.1 Subbands and effective mass

A cutting line that misses K by Δk_μ gives a subband

$$E_\mu(k_\parallel) = \pm\sqrt{\Delta_\mu^2 + (\hbar v_F k_\parallel)^2}, \qquad \Delta_\mu = \hbar v_F \Delta k_\mu$$

This is the dispersion of a relativistic particle with rest energy Δ_μ. In a semiconducting tube the lines nearest K and K′ miss by $1/3$, $2/3$, $4/3$, $5/3$, ... of the spacing, so the subband edges stand in the ratio 1: 2: 4: 5: .... In a metallic tube the first gapped subbands lie at 3 times Δ_1 of a semiconducting tube of the same diameter.

Near a band edge, $k_\parallel \ll \Delta k_\mu$, the hyperbola is close to a parabola, $E \approx \Delta_\mu + \hbar^2 k_\parallel^2/2m^*$, with

$$m^* = \frac{\Delta_\mu}{v_F^2}$$

For the lowest subband this gives $m^* = E_g/2v_F^2 \approx 0.09\,m_0/d(\text{nm})$.

> **Worked example: the effective mass**
>
> For $d = 1.5$ nm, $\Delta_1 = E_g/2 = 0.26$ eV $= 4.1 \times 10^{-20}$ J. Then $m^* = 4.1 \times 10^{-20}/(8.7 \times 10^5)^2 = 5.4 \times 10^{-32}$ kg $= 0.06\,m_0$, the same for electrons and holes. This is lighter than the conduction-band mass of silicon ($m_t = 0.19\,m_0$) and comparable with InGaAs. The symmetry between electrons and holes is a striking difference from silicon, where holes are much heavier. A nanotube can make n-type and p-type transistors of equal strength.

5.2 The density of states

In one dimension a subband contributes a density of states $D(E) \propto (\mathrm{d}E/\mathrm{d}k)^{-1}$, which diverges where the band is flat, at its edge. For the hyperbolic subband, per unit length and including spin and the two valleys K and K′:

$$D(E) = \frac{8}{hv_F}\frac{|E|}{\sqrt{E^2 - \Delta_\mu^2}} \qquad (|E| > \Delta_\mu)$$

summed over subbands. Each subband edge is a sharp peak, a **van Hove singularity**. In a metallic tube the linear bands give a constant density of states at the Fermi level, $D_0 = 8/hv_F \approx 2.2$ states eV^{-1} nm^{-1}: small, but not zero. That constant will reappear in Document 2 as the **quantum capacitance**, $C_Q = e^2 D_0 \approx 0.4$ fF/μm.

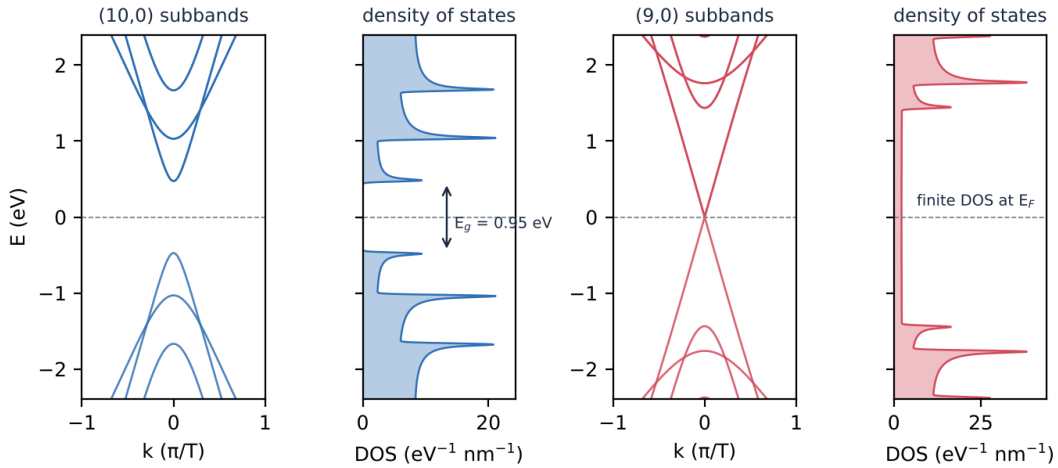


Figure 1.7. *Left pair: the subbands of a (10,0) tube (semiconducting, E_g = 0.95 eV) and its density of states. Right pair: the same for a (9,0) tube (metallic), with linear bands crossing at $E = 0$ and a constant density of states between the first peaks. The sharp peaks are van Hove singularities. Figure made for this course from the zone-folding model.*

5.3 Identifying tubes by their optical transitions

Light is absorbed most strongly between mirror-image van Hove peaks: E_{11} (first valence peak to first conduction peak), E_{22}, and so on. Plotting these energies against diameter for every (n, m) gives the **Kataura plot** (Kataura *et al.*, 1999), of which Figure 1.6 is a simple version. Because each (n, m) has its own (E_{11}, E_{22}) pair, a photoluminescence map, excitation at E_{22} and emission at E_{11}, identifies the species in a sample. This is how the purity of sorted material is measured (Section 7).

Check your understanding

1. Why do the subband edges of a semiconducting tube stand in the ratio 1 : 2 : 4 : 5?
2. Compute the effective mass for a 1.0 nm tube. How does it compare with silicon?

6 Transport

> **Key ideas**
>
> A tube with its lowest subband occupied has four conducting channels (two spins × two valleys), so its conductance is at most $4e^2/h$: a resistance of at least 6.45 kΩ.
>
> At low bias electrons travel about 1 μm before scattering off acoustic phonons. At high bias they emit optical phonons every 10–15 nm, which caps the current in a long tube at about 25 μA.
>
> That current through a 1.5 nm cross-section is a current density of about 10^9 A/cm^2, a thousand times what copper can carry.

6.1 The Landauer conductance

Lecture 2 showed that a perfect one-dimensional conductor has a resistance set by its number of channels. In a nanotube each subband at the Fermi level carries two spins and two valleys, so $M = 4$ channels:

$$G = \frac{4e^2}{h}\mathcal{T}, \qquad R_Q = \frac{h}{4e^2} = 6.45\ \text{k}\Omega$$

where $\mathcal{T} \leq 1$ is the transmission. A metallic tube has these four channels at any doping; a semiconducting tube has them once the gate fills its first subband. Either way **no single nanotube can have a resistance below 6.45 kΩ**, however short it is and however good its contacts. This *contact* resistance is not a defect of processing. It is the resistance of fitting a three-dimensional metal onto four one-dimensional channels, and it is the largest single obstacle to CNT transistors (Document 2).

With scattering, the transmission of a tube of length L and mean free path λ is $\mathcal{T} = \lambda/(\lambda + L)$, so

$$R = \frac{h}{4e^2}\left(1 + \frac{L}{\lambda}\right)$$

Short tubes ($L \ll \lambda$) are **ballistic** and show $R \approx 6.45$ kΩ; long tubes are diffusive, with R growing linearly with length.

6.2 Scattering and current saturation

The sp^2 lattice is very stiff, and there are no dopants in the tube, so the main scatterers at room temperature are phonons:

- **Acoustic phonons** scatter weakly. At low bias the mean free path is about 0.5–1 μm in good tubes, and the low-field mobility can exceed 10^5 cm^2/V s (Dürkop *et al.*, 2004), about a hundred times that of silicon. Mobility rises roughly as d^2, another reason to prefer wider tubes.
- **Optical and zone-boundary phonons** have energies of about 0.16–0.20 eV. Once an electron gains that much energy from the field it emits one almost at once. At high bias the mean free path falls to 10–15 nm.

In a long tube, therefore, the current saturates at about 20–25 μA per tube (Yao, Kane and Dekker, 2000): every electron that gains $\hbar\Omega \approx 0.16$ eV emits a phonon and starts again. In a tube shorter than the optical-phonon length, electrons can cross before they have gained

enough energy, and currents of 60–70 μA per tube have been measured (Javey *et al.*, 2004). For a transistor the relevant figure is the short-tube one.

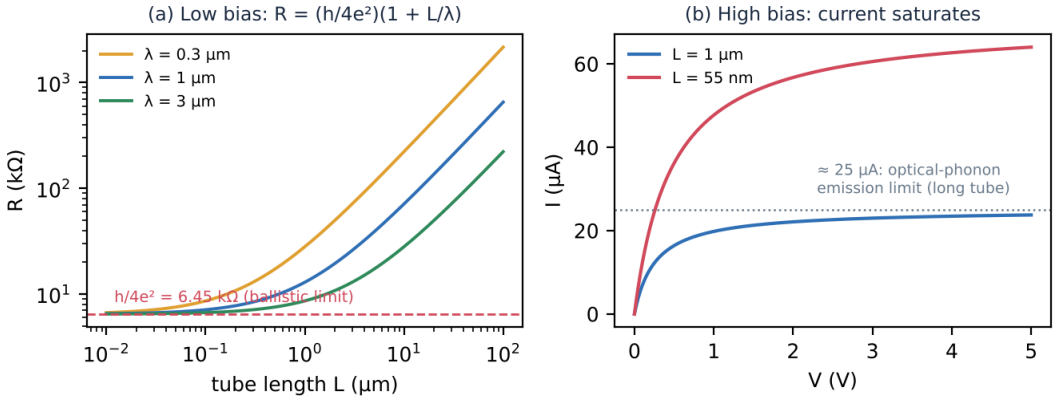


Figure 1.8. *(a) Low-bias resistance of a single tube against length for three mean free paths: ballistic at 6.45 kΩ for short tubes, rising linearly for long ones. (b) Current–voltage curves of a long and a short metallic tube, from the simple saturation model $I = V/(R_0 + V/I_0)$ with I_0 = 25 μA and 70 μA. Figure made for this course; illustrative values from the measurements cited in the text.*

> **Worked example: current density**
>
> A 1.5 nm tube has a cross-section of $\pi(0.75\ \text{nm})^2 = 1.8 \times 10^{-14}\ \text{cm}^2$. Carrying 25 μA, its current density is $25 \times 10^{-6} / 1.8 \times 10^{-14} = 1.4 \times 10^{9}\ \text{A/cm}^2$. Copper interconnects are limited by electromigration to about 10^6–$10^7\ \text{A/cm}^2$. The strong σ bonds hold the carbon atoms in place where copper atoms would be pushed along by the electron wind. Strictly, the area of a one-atom-thick tube is ill-defined, but every sensible choice gives a current density far beyond copper's.

6.3 Thermal and mechanical properties

The same stiff bonds make nanotubes excellent heat conductors. The thermal conductivity of a single SWNT at room temperature is about 3000–3500 W/m K (Pop *et al.*, 2006), compared with 400 for copper and 150 for silicon. Heat travels along the axis, not across it. The Young's modulus is about 1 TPa, five times that of steel, and the tensile strength is tens of GPa.

> **A correction to the old notes.** The Springer chapter in the original L4 material states, under *thermal conductivity*, that "CNTs can exhibit superconductivity below 20 K due to the strong in-plane C–C bonds". Those are two unrelated things. Superconductivity has been reported in very thin (0.4 nm) tubes inside zeolite channels at about 15 K, and in ropes at about 0.5 K. It has nothing to do with heat conduction or bond strength, and it is irrelevant to nanotube electronics.

Check your understanding

1. Why can a nanotube transistor never have a total resistance below 6.45 kΩ, however short its channel?
2. Why does the saturation current of a short tube exceed that of a long one?

7 From physics to material: growth, sorting and alignment

> **Key ideas**
>
> No growth method yet makes a single (n, m) at scale. Material for electronics is grown, then sorted in solution, then deposited as aligned arrays.
>
> By 2020, sorted and aligned arrays reached more than 99.9999% semiconducting tubes at 100–200 tubes per µm on 10 cm wafers.
>
> What is still missing is control of pitch, diameter and placement good enough for billions of identical transistors.

Lecture 2, Document 3 described the three ways of growing nanotubes: arc discharge, laser ablation and chemical vapour deposition (CVD). This section adds what matters for transistors.

7.1 Growth

In CVD a hydrocarbon (methane, ethylene, ethanol) decomposes on nanoparticles of iron, cobalt or nickel at 600–1000 °C; carbon dissolves in the particle and precipitates as a tube whose diameter follows the particle's. CVD can grow tubes directly on a wafer. Lined up along the atomic steps of quartz or sapphire, they grow as parallel arrays up to millimetres long, and a single tube has been grown to 55 cm (Zhang *et al.*, 2013).

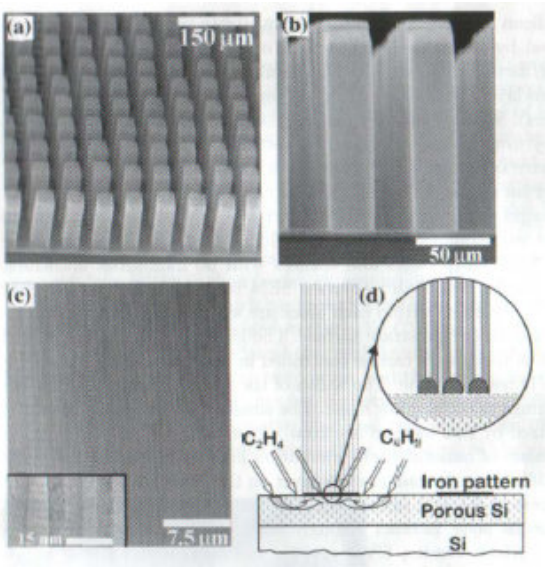


Figure 1.9. *Vertically aligned multi-walled nanotubes grown by CVD from a patterned iron catalyst on porous silicon: (a, b) SEM images of the "towers", which align themselves through van der Waals forces between neighbouring tubes; (c) close-up; (d) the base-growth process, with the catalyst staying on the substrate. Source: S. Fan et al., Science 283, 512 (1999); reproduced from S. Rai, University of Notre Dame lecture slides. The original slide gives the carbon source as methane, but the figure shows ethylene (C_2H_4), which is what Fan et al. used.*

Growing a single chirality directly has been demonstrated. Using a solid tungsten–cobalt alloy catalyst whose atomic structure matches the end of a (12,6) tube, Yang *et al.* (2014) grew samples that were about 92% (12,6). But the yield and wafer coverage are far from what manufacturing needs, and directly grown arrays still contain a few per cent of metallic tubes. Every CNT technology used for circuits in 2026 starts from sorted tubes.

7.2 Sorting

Sorting works on tubes dispersed in solution:

- **Density-gradient ultracentrifugation** (Arnold *et al.*, 2006). Surfactant coating differs slightly between metallic and semiconducting tubes, and between diameters, so they settle at different depths.
- **Gel chromatography** (Liu *et al.*, 2011). Semiconducting tubes stick to an agarose gel more strongly than metallic ones, and individual (n, m) species elute separately.
- **Conjugated-polymer wrapping** (Nish *et al.*, 2007). Polymers such as polyfluorenes wrap selectively around semiconducting tubes of a chosen diameter range and carry them into an organic solvent, leaving metallic tubes behind. Repeated, this gives the highest purities, and it is the basis of most aligned-array work today.

Purity is quoted as the fraction of semiconducting tubes. For VLSI the target is usually put at 99.9999% or better. Even then a chip with 10^{10} tubes contains about 10^4 metallic ones, so circuits must also tolerate them (Document 2).

7.3 Alignment and density

A transistor needs many tubes in parallel to carry enough current: the target is a pitch of about 5 nm, i.e. 100–200 tubes per µm of width, all parallel. In 2020 Liu *et al.* (Peking University) used polymer-sorted tubes and a **dimension-limited self-alignment** process to make wafer-scale arrays on 10 cm substrates with 100–200 tubes per µm and an estimated semiconducting purity above 99.9999%. Transistors made from these arrays reached on-currents above 1 mA/µm. This result, and others like it, moved the field from "can it be made at all?" to "can it be made uniformly?" Document 2 picks up from there.

Check your understanding

1. Why is direct growth not yet used for CNT circuits?
2. A chip has 10^9 transistors, each with 20 tubes, from material of 99.9999% purity. Roughly how many metallic tubes does it contain?

8 Summary

- Graphene's π bands, from one-parameter tight binding, touch at the K points; near K the dispersion is a cone with $\hbar v_F = 3/2\, a_{cc}\gamma_0$.
- A tube is labelled by $\mathbf{C}_h = n\mathbf{a}_1 + m\mathbf{a}_2$: $d = (a/\pi)\sqrt{n^2 + nm + m^2}$, chiral angle 0–30°.
- The boundary condition $\mathbf{k} \cdot \mathbf{C}_h = 2\pi\mu$ leaves cutting lines $2/d$ apart. A tube is metallic if $(n - m)$ mod $3 = 0$, otherwise semiconducting with $E_g = 2a_{cc}\gamma_0/d \approx 0.77$ eV nm$/d$.
- Subbands are hyperbolic, with $m^* \approx 0.09\, m_0/d$ (nm) for electrons and holes alike, and a density of states with van Hove peaks.
- Four channels give a minimum resistance of $h/4e^2 = 6.45$ kΩ per tube. The low-bias mean free path is about 1 µm; at high bias, optical phonons cap the current at about 25 µA per long tube.
- Electronic-grade material is sorted (to >99.9999% semiconducting) and aligned (100–200 tubes per µm); uniformity, not purity, is now the main problem.

Corrections to the old notes

Old note	What it said	Correction
Kaushik & Majumder, ch. 2	Metallic condition $n_1 - n_2 = 3i$ stated for armchair tubes	The rule holds for every (n, m); armchair tubes are a special case
Kaushik & Majumder, ch. 2	Eq. 2.17: $m = 2q + 1$ in the text, $2q - 1$ in the equation	For $n = 3q + 1$ the nearest line is $m = 2q + 1$; the equation has a typo
Kaushik & Majumder, ch. 2	"Superconductivity below 20 K due to the strong C–C bonds", under thermal conductivity	Unrelated effects; the thermal conductivity is about 3000 W/m K (Section 6.3)
Kaushik & Majumder, ch. 2	Nanotubes "up to 18 cm" long	55 cm (Zhang *et al.*, 2013)
Kaushik & Majumder; L2	$E_g \approx 0.7/d$ (old note) vs $0.8/d$ (L2)	Both are $2a_{cc}\gamma_0/d$ with different γ_0; this lecture uses 0.77
Dandeliya & Srivastava	"a = 0.142 Å, bond length"	a_{cc} = 1.42 Å = 0.142 nm; the lattice vectors use a = 2.46 Å
Rai (Notre Dame)	CVD towers grown from methane	The figure (Fan *et al.*, 1999) shows ethylene

Words to know

Term	Meaning
Tight binding	A band model built from atomic orbitals and the energy for an electron to hop between neighbours
γ_0	The nearest-neighbour hopping energy of graphene, about 2.7 eV
K point	A corner of graphene's Brillouin zone, where the π and π* bands touch
Dirac cone	The conical dispersion $E = \pm\hbar v_F \kappa$ near K
Chiral vector	$\mathbf{C}_h = n\mathbf{a}_1 + m\mathbf{a}_2$, the circumference of the tube
Zone folding	Sampling the graphene bands along the lines allowed by the circumferential boundary condition
van Hove singularity	A peak in the 1D density of states at a subband edge
Kataura plot	Optical transition energies of all tubes against diameter
Quantum capacitance	e^2 times the density of states: the capacitance of filling the tube's own states

In-class animation

Nanotube band structure (faizverse.github.io/nanoelectronic-devices/L4). Choose (n, m) and watch the cutting lines move across the Brillouin zone. The page draws the subbands and the density of states and reports d, θ, E_g and whether the tube is metallic. Try (9,0), (10,0) and (11,0), then a series of tubes with the same $n - m$.

Sources and further reading

Textbooks and reviews

- R. Saito, G. Dresselhaus and M. S. Dresselhaus, *Physical Properties of Carbon Nanotubes*, Imperial College Press, 1998. *The standard source for zone folding and the notation used here.*
- A. Javey and J. Kong (eds.), *Carbon Nanotube Electronics*, Springer, 2009.
- P. Avouris, Z. Chen and V. Perebeinos, "Carbon-based electronics", *Nature Nanotechnology* 2, 605 (2007).
- B. K. Kaushik and M. K. Majumder, *Carbon Nanotube Based VLSI Interconnects*, SpringerBriefs, 2015, ch. 2 (in the old notes).

Papers cited

- S. Iijima, "Helical microtubules of graphitic carbon", *Nature* 354, 56 (1991).
- H. Kataura *et al.*, "Optical properties of single-wall carbon nanotubes", *Synthetic Metals* 103, 2555 (1999).
- Z. Yao, C. L. Kane and C. Dekker, "High-field electrical transport in single-wall carbon nanotubes", *Phys. Rev. Lett.* 84, 2941 (2000).
- T. Dürkop *et al.*, "Extraordinary mobility in semiconducting carbon nanotubes", *Nano Lett.* 4, 35 (2004).
- A. Javey *et al.*, "High-field quasiballistic transport in short carbon nanotubes", *Phys. Rev. Lett.* 92, 106804 (2004).
- E. Pop *et al.*, "Thermal conductance of an individual single-wall carbon nanotube above room temperature", *Nano Lett.* 6, 96 (2006).
- M. S. Arnold *et al.*, "Sorting carbon nanotubes by electronic structure using density differentiation", *Nature Nanotechnology* 1, 60 (2006).
- A. Nish *et al.*, "Highly selective dispersion of single-walled carbon nanotubes using aromatic polymers", *Nature Nanotechnology* 2, 640 (2007).
- H. Liu *et al.*, "Large-scale single-chirality separation of single-wall carbon nanotubes by simple gel chromatography", *Nature Communications* 2, 309 (2011).
- R. Zhang *et al.*, "Growth of half-meter long carbon nanotubes based on Schulz–Flory distribution", *ACS Nano* 7, 6156 (2013).
- F. Yang *et al.*, "Chirality-specific growth of single-walled carbon nanotubes on solid alloy catalysts", *Nature* 510, 522 (2014).
- L. Liu *et al.*, "Aligned, high-density semiconducting carbon nanotube arrays for high-performance electronics", *Science* 368, 850 (2020).

From the original L4 notes

- S. Rai, *Carbon Nanotubes*, University of Notre Dame lecture slides.
- Kaushik and Majumder (above).